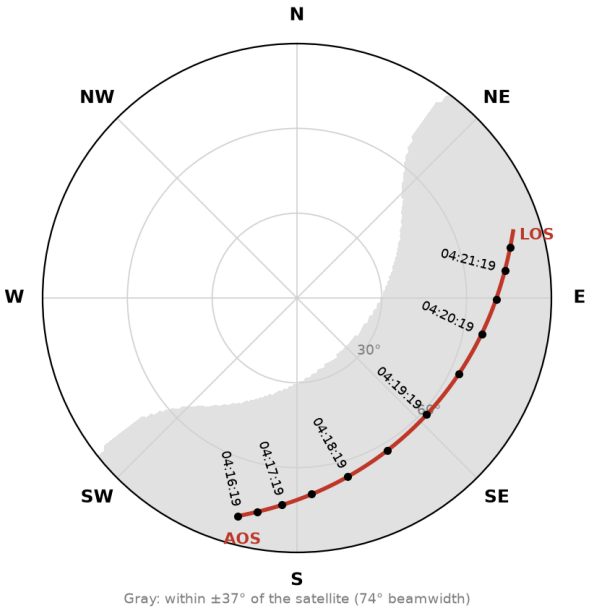
ISS Pass Plan

Tuesday 10/20/2026 Rise 04:16:19 Max 04:19:16 Set 04:22:14 Max El 28° Az 195°→73°

TX (uplink) 145.990 MHz FM, 67.0 Hz CTCSS **RX (downlink) 437.800 MHz FM** — Doppler-corrected, rounded to 5 kHz (error up to 2.5 kHz)

Time	TX MHz	RX MHz	Az	El
04:16:19	145.985	437.810	195°	10°
04:16:49	145.985	437.810	190°	13°
04:17:19	145.990	437.805	184°	17°
04:17:49	145.990	437.805	176°	20°
04:18:19	145.990	437.805	164°	24°
04:18:49	145.990	437.800	149°	27°
04:19:19	145.990	437.800	132°	28°
04:19:49	145.990	437.795	115°	27°
04:20:19	145.990	437.795	101°	23°
04:20:49	145.990	437.795	90°	20°
04:21:19	145.995	437.790	82°	16°
04:21:49	145.995	437.790	77°	12°



Gray: within ±37° of the satellite (74° beamwidth)

Yaesu VX-6R + speaker mic: uplink TX

- **Set Mode access:** [F/W] then [0(SET)], dial to the item, [0(SET)] to edit, dial = value, PTT = save
- **Frequency:** VHF band, VFO: key in digits for 145.990; the dial retunes for Doppler
- **Tone:** [F/W] then [MODE(SPS) SQ TYP], dial to TONE (not T SQL), PTT to save; [F/W] then [2(CODE)], dial to 67.0 Hz, press [2(CODE)] to save
- **Power:** [F/W] then [7(LK)TXPO] cycles HIGH / LOW3 (2.5 W) / LOW2 (1 W) / LOW1; use LOW3
- **Set Mode 1 APO:** OFF
- **Set Mode 4 ARS:** OFF
- **Set Mode 6 BCLO:** OFF, so PTT is never blocked by the downlink
- **Set Mode 27 HLF.DEV:** OFF: full deviation
- **Set Mode 37 MCGAIN:** LVL 5, adjust to your voice on the speaker mic
- **Set Mode 51 RPT:** SIMP (simplex); shortcut [F/W] [6(RPT)]
- **Set Mode 67 TOT:** 1 min (default 3)
- **Set Mode 70 TXSAVE:** OFF
- **Do not:** tap [7(LK)TXPO] alone: it turns on the Internet DTMF beep at every TX

Alinco DJ-VX50: downlink RX

- **Menu access:** FUNC, then 8, UP/DOWN to the menu number, dial = value, FUNC = confirm, ESC = exit
- **Frequency:** VFO mode: key in digits, e.g. 437.800 = 4 3 7 8 0 0; the dial then retunes for Doppler
- **Menu 01 R-CDC:** OFF (default): a tone squelch would mute the downlink
- **Menu 03 STEP:** 5K (default is 10K), so each dial click is one ladder step
- **Menu 04 W/N:** WIDE (default)
- **Menu 17 SAVE:** OFF (default is ON): battery saver can clip weak openings
- **Menu 19 SQL:** start at 1-2 (default 9 is tight); raise only if noise is constant
- **Menu 20 REPEAT:** OFF (default), and FUNC 4 until no offset is shown: simplex
- **Keypad lock:** FUNC, then hold #; the dial still retunes, and PTT stays untouched on this radio
- **Audio:** Earphone, volume up enough to hear a weak FM carrier

Notes

- ISS FM repeater availability varies (check the AMSAT / ARISS status pages).
- If the crew is using the station radio for other modes it will be silent.

General

- **Identify:** KF8EEZ, short calls; low power. Keep both radios and the TX antenna well away from the SDR
- **Two radios:** Yaesu VX-6R + MH-57A4B speaker mic = uplink TX (log-periodic); Alinco DJ-VX50 = downlink RX. Use an earphone on the Alinco so its speaker doesn't feed the speaker mic
- **Aim (log-periodic):** point AT satellite: pre-aim at rise Az, follow the Az/El ladder; slant 45°